

Random column subsets of fixed-sparsity matrices: an unrestricted theorem

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Abstract

For every fixed column sparsity, aspect ratio, and positive singular-value threshold, we prove that a uniformly sampled proportional-size column subset of any sufficiently redundant signed sparse dictionary has a positive fraction of singular values below that threshold, except with exponentially small probability. No hypothesis is imposed on intersections of column supports. The argument combines a covariance polynomial filter, finite signed conditional trajectories, and a common reservoir of sampled columns. A simultaneous-witness lemma converts reconstruction events into a lower bound for an eigenvalue count. A collision coupling transfers its expectation to sampling without replacement, after which a transposition martingale supplies concentration. This gives an affirmative answer to the exact target TR-07 in OpenProblemsInNLA, corresponding to the nontrivial regime of Huang–Rudelson–Tikhomirov’s Conjecture 7.1. The proof is analytic; no finite computation is a premise.

1 Statement and explicit constants

For integers $k \geq s \geq 1$, write

$$\mathcal{V}_{k,s} = \{u \in \{-1, 0, 1\}^k : |\text{supp } u| = s\}.$$

For a real $k \times r$ matrix A and $\eta > 0$, set

$$N_\eta(A) = \#\{j \in \{1, \dots, r\} : \lambda_j(A^T A) \leq \eta^2\},$$

where the eigenvalues are counted with multiplicity, including zeros. Thus, when $r \leq k$, this is the number of its r singular values at most η . Throughout, eigenvalues of a symmetric matrix are ordered increasingly.

Theorem 1.1 (Uniform finite statement). *Fix $s \geq 1$, $0 < \beta \leq 1$, and $\eta > 0$. There are constants $0 < \rho < 1$ and $k_0 \in \mathbb{N}$, depending only on (s, β, η) , with the following property. If*

$$k \geq k_0, \quad \beta k \leq r \leq k, \quad n \geq r/\rho,$$

if $M \in \{-1, 0, 1\}^{k \times n}$ has exactly s nonzero entries per column, and if I is a uniform r -element subset of $\{1, \dots, n\}$, then

$$\mathbb{P}\{N_\eta(M_{:I}) < \rho r/4\} \leq \exp(-\rho^2 r/8). \quad (1)$$

The constants are uniform over every choice of signs and column supports.

Here is one explicit, very inefficient choice. Put

$$\begin{aligned} \varepsilon &= \eta/2, & L &= \left\lceil \frac{4s^2}{\varepsilon^2} \right\rceil, & S_L &= 2^L - 1, \\ b &= \left\lceil \frac{8sS_L^2}{\varepsilon^2} \right\rceil, & J_0 &= bL, & a &= \frac{\beta}{16J_0 + \beta}, \\ \rho &= \frac{3}{32}a^{J_0}, & k_0 &= \max \left\{ s, \left\lceil \frac{4J_0}{\beta} \right\rceil \right\}. \end{aligned} \tag{2}$$

All these quantities are finite and positive where appropriate, and are chosen before k, r, n , or M . In particular, $r \geq 4J_0 \geq 4$ under the theorem's hypotheses.

Corollary 1.2 (TR-07). *Fix an integer $s \geq 2$ and a finite constant $C \geq 1$. Suppose that $r_k \leq k$, $k/r_k \rightarrow C$, and $n_k/r_k \rightarrow \infty$. For every deterministic sequence*

$$M_k \in \{-1, 0, 1\}^{k \times n_k}$$

with exactly s nonzero entries per column, and a uniform r_k -element column set I_k , one has, for every $\eta > 0$,

$$\mathbb{P} \left\{ \inf_{\|x\|_2=1} \|(M_k)_{:I_k} x\|_2 > \eta \right\} \rightarrow 0.$$

There is no condition on intersections of column supports.

Deduction from Theorem 1.1. Fix $\eta > 0$ and use $\beta = 1/(2C)$. For all sufficiently large k , one has $r_k \geq \beta k$, $k \geq k_0$, $n_k \geq r_k/\rho$, and $\rho r_k/4 \geq 1$. The event in the corollary is then contained in $\{N_\eta((M_k)_{:I_k}) < \rho r_k/4\}$. Its probability is at most $\exp(-\rho^2 r_k/8)$, which tends to zero. $\square$

Corollary 1.2 is the canonical statement of TR-07 [2]. It is the $r_k \leq k$ regime of Conjecture 7.1 in [1]. If the latter is considered with $r_k > k$, the infimum definition of the least singular value is zero by dimension; the other indices are covered by the same argument. The proof below does not invoke the support-overlap theorem of [1].

2 Two deterministic facts about small singular values

Lemma 2.1 (Simultaneous witnesses). *Suppose a matrix A contains m designated center columns and a disjoint set of reservoir columns. Suppose each center can be approximated to Euclidean error at most $\varepsilon = \eta/2$ by a linear combination of reservoir columns. Then*

$$N_\eta(A) \geq m/2.$$

The linear combinations may share reservoir columns and may have arbitrarily large coefficients.

Proof. The assertion is immediate for $m = 0$. Otherwise, collect the m dependence coefficient vectors into a matrix V , with a coefficient 1 at the corresponding center and minus the reconstruction coefficients at the reservoir positions. After permuting column coordinates, the block of V at the selected centers is I_m , its reservoir block is $-X$, and all other rows vanish. Therefore

$$V^T V = I_m + X^T X \succeq I_m, \quad \|AV\|_F^2 \leq m\varepsilon^2.$$

Set $Q = V(V^T V)^{-1/2}$. Then $Q^T Q = I_m$, and

$$\|AQ\|_F^2 \leq \|AV\|_F^2 \leq m\eta^2/4.$$

The minimum principle for sums of eigenvalues gives

$$\sum_{j=1}^m \lambda_j(A^T A) \leq \text{tr}(Q^T A^T A Q) \leq m\eta^2/4. \quad (3)$$

To see the minimum principle directly, diagonalize $A^T A$ and write its trace against the orthogonal projection QQ^T as $\sum_j \lambda_j w_j$, where $0 \leq w_j \leq 1$ and $\sum_j w_j = m$. This is minimized by assigning weight one to the first m eigenvalues. If $N_\eta(A) < m/2$, more than $m/2$ of the first m eigenvalues exceed η^2 , contradicting (3). $\square$

Lemma 2.2 (Column replacement is one-Lipschitz). *If two matrices with the same number of columns differ in at most h column positions, then*

$$|N_\eta(A) - N_\eta(B)| \leq h.$$

Also, N_η is invariant under column permutations.

Proof. Deleting a column replaces the Gram matrix by a principal submatrix. The interlacing inequalities imply that the count in $(-\infty, \eta^2]$ can decrease by either zero or one. If A and B share all but h columns, delete the exceptional positions from both to obtain a common matrix C . Each of $N_\eta(A)$ and $N_\eta(B)$ belongs to $[N_\eta(C), N_\eta(C) + h]$. Permuting columns conjugates the Gram matrix by a permutation matrix. $\square$

3 A finite reconstruction using conditional trajectories

Lemma 3.1 (Pruning). *For every probability law μ on $\mathcal{V}_{k,s}$ there is a subset G of its support with $g := \mu(G) \geq 3/4$ such that every active coordinate satisfies*

$$\mu\{u \in G : u_i \neq 0\} \geq \frac{1}{4k}. \quad (4)$$

Here active means that the quantity on the left is nonzero.

Proof. Begin with the full support. If a coordinate has positive remaining incident mass less than $1/(4k)$, delete every remaining vector incident to that coordinate. Always measure mass using the original, unnormalized law μ . Once charged, a coordinate has no incident vectors left and is never charged again. There are at most k charges, each less than $1/(4k)$. Thus at most $1/4$ of the total mass is removed, and the terminal set has the asserted property. $\square$

Let $\nu = \mu(\cdot \mid G)$. Restrict all covariance operators below to the active coordinate subspace, and write

$$p_i = \nu\{u_i \neq 0\}, \quad D = \text{diag}(p_i), \quad \Sigma = \mathbb{E}_\nu uu^T, \quad T = \Sigma D^{-1}.$$

Only D is inverted. It is positive definite on this subspace; Σ is allowed to be singular. Zero extension to the original k coordinates leaves every vector norm unchanged. Since every vector has s nonzero entries,

$$\text{tr } D = s. \quad (5)$$

Lemma 3.2 (Averaged covariance filter). *For every integer $L \geq 1$,*

$$\mathbb{E}_\nu \left\| (I - T/s)^L u \right\|_2^2 \leq \frac{s^2}{2L + 1}. \quad (6)$$

Proof. For every deterministic $u \in \mathcal{V}_{k,s}$ and every x ,

$$(u^T x)^2 \leq s \sum_{i \in \text{supp } u} x_i^2.$$

Averaging gives $\Sigma \preceq sD$. Consequently

$$R = D^{-1/2} \Sigma D^{-1/2}, \quad 0 \preceq R \preceq sI, \quad T = D^{1/2} R D^{-1/2}.$$

Direct substitution and cyclicity of trace yield

$$\begin{aligned} \mathbb{E}_\nu \left\| (I - T/s)^L u \right\|_2^2 &= \text{tr} \left((I - T/s)^L \Sigma ((I - T/s)^L)^T \right) \\ &= \text{tr} (DR(I - R/s)^{2L}). \end{aligned}$$

For $0 \leq \lambda \leq s$, differentiation, or maximization of $t(1-t)^{2L}$ on $[0, 1]$, gives

$$0 \leq \lambda(1 - \lambda/s)^{2L} \leq \frac{s}{2L+1}.$$

The spectral theorem and (5) prove (6). In particular, no Euclidean operator-norm contraction of the generally nonnormal matrix $I - T/s$ is being assumed. $\square$

A signed conditional transition starts from a state v that is a signed copy of a vector in G . Choose i uniformly from its s support coordinates, sample w from ν conditional on $w_i \neq 0$, and replace v by $v_i w_i w$. The new state is again a signed copy of a vector in G , has norm $\sqrt{s}$, and satisfies

$$\mathbb{E}[v_i w_i w \mid v] = \frac{1}{s} \sum_{i \in \text{supp } v} \frac{v_i}{p_i} \mathbb{E}_\nu[u_i u] = \frac{T}{s} v. \quad (7)$$

The sign of a state is carried in the coefficients; the law ν need not be sign symmetric.

Lemma 3.3 (Finite ideal reconstruction). *Use ε, L, b, J_0 from (2). Given $u \sim \nu$, run b independent length- L conditional trajectories from u , independent conditional on u . There is a linear combination Y of their $J_0 = bL$ visited sample vectors such that*

$$\mathbb{P}_{\text{ideal}} \{ \|u - Y\|_2 \leq \varepsilon \} \geq 3/4. \quad (8)$$

The probability averages over both the center and the trajectories.

Proof. Let $v_{h,j}$ denote the state after step j of trajectory h , with $v_{h,0} = u$. Iteration of (7) gives

$$\mathbb{E}[v_{h,j} \mid u] = (T/s)^j u.$$

For $c_j = (-1)^{j+1} \binom{L}{j}$, put

$$Y_h = \sum_{j=1}^L c_j v_{h,j}, \quad Y = \frac{1}{b} \sum_{h=1}^b Y_h.$$

Each signed state is a signed sampled vector, so Y belongs to the claimed span. The binomial identity gives

$$\mathbb{E}[Y \mid u] = [I - (I - T/s)^L] u.$$

Also $\|Y_h\|_2 \leq \sqrt{s} \sum_j |c_j| = \sqrt{s} S_L$. Conditional independence of the trajectories, not of successive states within a trajectory, gives

$$\mathbb{E} \|Y - \mathbb{E}[Y \mid u]\|_2^2 \leq \frac{s S_L^2}{b}.$$

The conditional centered error is orthogonal in mean square to the bias. Lemma 3.2 therefore implies

$$\mathbb{E} \|u - Y\|_2^2 \leq \frac{s^2}{2L+1} + \frac{s S_L^2}{b} \leq \frac{\varepsilon^2}{4}. \quad (9)$$

Markov's inequality proves (8). This construction uses exactly bL transitions; separate trajectories for each of the different powers are unnecessary. $\square$

4 Finding reconstructions in an independent sample

Proposition 4.1 (Uniform expectation bound). *Let μ be any law on $\mathcal{V}_{k,s}$. Let A have r independent columns with law μ , where $k \geq k_0$ and $\beta k \leq r \leq k$. Then*

$$\mathbb{E}N_\eta(A) \geq \rho r, \quad (10)$$

with ρ from (2).

Proof. Use Lemma 3.1. Designate the first $m = \lfloor r/2 \rfloor$ columns as centers and the remaining columns as a reservoir. Partition the first $J_0\ell$ reservoir columns into J_0 fixed disjoint chunks of size

$$\ell = \left\lfloor \frac{r-m}{J_0} \right\rfloor \geq \frac{r}{4J_0},$$

since $r \geq 4J_0$. For any one center, declare failure if it is outside G . Otherwise attempt the b trajectories of Lemma 3.3 in a fixed order, using one fresh chunk for each transition. Once its requested coordinate i has been chosen, take the first vector w in that chunk satisfying $w \in G$ and $w_i \neq 0$. Declare failure if none exists.

Write $\pi_i = \mu\{w \in G : w_i \neq 0\} = gp_i$. Every requested coordinate is active, so $\pi_i \geq 1/(4k)$. Its chunk-hit probability is

$$h_i = 1 - (1 - \pi_i)^\ell \geq \frac{\ell\pi_i}{1 + \ell\pi_i} \geq \frac{\beta}{16J_0 + \beta} = a.$$

For the first inequality, $(1 - p)^\ell \leq (1 + \ell p)^{-1}$ for $0 \leq p \leq 1$: for $p < 1$ use $(1 - p)^{-\ell} \geq 1 + \ell p/(1 - p) \geq 1 + \ell p$, and for $p = 1$ the assertion is immediate.

For any set B of vectors, the first-hit formula is exactly

$$\mathbb{P}\{\text{hit, selected vector} \in B \mid i\} = h_i \nu(B \mid w_i \neq 0). \quad (11)$$

Indeed, summing the probabilities that positions $1, \dots, \ell$ supply the first acceptable vector gives this identity. Given the past for this particular center, its next chunk is an independent sequence with law μ . Thus each successful transition kernel, including the uniform support-coordinate choice, is at least a times the ideal transition kernel. Composing these nonnegative kernels over all J_0 transitions proves that the law of completed trajectories dominates a^{J_0} times the ideal law.

The center itself has law $g\nu$ when restricted to G . Therefore, by (8), the probability that this center produces a completed reconstruction of error at most ε is at least

$$ga^{J_0} \cdot \frac{3}{4} \geq \frac{9}{16}a^{J_0}. \quad (12)$$

This is domination of the unnormalized successful law. Conditioning on all hits need not preserve the ideal trajectory law; that incorrect assertion is not used.

Reuse the same reservoir chunks for all centers. Analyze each center's marginal probability as above, without conditioning on any other center's outcomes. Let Z be the number of successes, with their auxiliary coordinate-choice randomness included. Linearity of expectation and (12) give

$$\mathbb{E}Z \geq \frac{9}{16}a^{J_0}m.$$

For every realization, all successful centers are reconstructed using only the common reservoir, which is disjoint from every center. Lemma 2.1 applies simultaneously, so $N_\eta(A) \geq Z/2$. No independence among the center-success events is needed. Since $m \geq r/3$,

$$\mathbb{E}N_\eta(A) \geq \frac{1}{2}\mathbb{E}Z \geq \frac{3}{32}a^{J_0}r = \rho r.$$

The auxiliary randomness has disappeared from the left side, which depends only on the actual r sampled columns. $\square$

5 Sampling without replacement and concentration

Lemma 5.1 (Expectation comparison). *Let M be any real matrix with n columns and let $r \leq n$. If $J_1, \dots, J_r$ are independent uniform column indices and I is a uniform r -element subset, then*

$$\mathbb{E}N_\eta(M_{:I}) \geq \mathbb{E}N_\eta(M_{:J}) - \frac{r(r-1)}{2n}. \quad (13)$$

Repeated indices in $M_{:J}$ mean repeated columns, not a set.

Proof. Construct distinct indices $I_1, \dots, I_r$ in order. Keep $I_t = J_t$ if J_t is unused among $I_1, \dots, I_{t-1}$; otherwise choose I_t uniformly among unused indices, using fresh randomness. Given the entire past, for each unused i ,

$$\mathbb{P}\{I_t = i \mid \text{past}\} = \frac{1}{n} + \frac{t-1}{n} \frac{1}{n-t+1} = \frac{1}{n-t+1}.$$

Thus (I_t) is uniformly sampled without replacement. If $H = \#\{t : I_t \neq J_t\}$, then

$$\mathbb{E}H = \sum_{t=1}^r \frac{t-1}{n} = \frac{r(r-1)}{2n}.$$

Lemma 2.2 gives $N_\eta(M_{:I}) \geq N_\eta(M_{:J}) - H$ in every realization. Take expectations. $\square$

Lemma 5.2 (Concentration for a uniform subset). *Let F be a real function of r -element subsets of $\{1, \dots, n\}$. Suppose $|F(S) - F(S')| \leq 1$ whenever S, S' differ by exchanging one element. If I is a uniform r -element subset, then for $t > 0$,*

$$\mathbb{P}\{F(I) - \mathbb{E}F(I) \leq -t\} \leq \exp(-2t^2/r). \quad (14)$$

Proof. Reveal a uniformly ordered sample $I_1, \dots, I_r$ without replacement, and use its Doob martingale. Fix a revealed prefix before time j and two possible next values a, b . Couple uniform completions after a and after b by transposing the labels a, b in the remaining sample. If the completion after a contains b , the resulting final subsets agree. Otherwise they differ by exchanging a and b . The corresponding conditional expectations of F differ by at most one. Hence each martingale increment, conditional on the past, is centered and belongs to an interval of length at most one.

For completeness, a centered bounded random variable X in an interval of length one obeys $\mathbb{E}e^{\lambda X} \leq e^{\lambda^2/8}$. Indeed, if $\phi(\lambda) = \log \mathbb{E}e^{\lambda X}$, then $\phi(0) = \phi'(0) = 0$ and $\phi''(\lambda)$ is a variance under the exponentially tilted law, bounded by $1/4$ because its support still lies in that interval. Integrating twice gives the bound. Applying it conditionally and iterating yields

$$\mathbb{E}e^{-\lambda(F(I) - \mathbb{E}F(I))} \leq e^{r\lambda^2/8}.$$

Exponential Markov and $\lambda = 4t/r$ prove (14). $\square$

Proof of Theorem 1.1. Give the n column indices of M equal mass and let μ be the resulting distribution of column vectors. It is a probability law on $\mathcal{V}_{k,s}$, irrespective of repeated column values. Proposition 4.1 and Lemma 5.1 imply

$$\mathbb{E}N_\eta(M_{:I}) \geq \rho r - \frac{r(r-1)}{2n} \geq \rho r/2,$$

because $n \geq r/\rho$. By Lemma 2.2, the function $F(I) = N_\eta(M_{:I})$ satisfies the hypothesis of Lemma 5.2. Apply that lemma with $t = \rho r/4$. The result is

$$\mathbb{P}\{N_\eta(M_{:I}) < \rho r/4\} \leq \exp\left(-\frac{2(\rho r/4)^2}{r}\right) = \exp(-\rho^2 r/8),$$

as claimed. $\square$

Scope, provenance, and verification status

The theorem covers every fixed s , including $s = 1$, with arbitrary signs, repeated columns, and unrestricted support intersections. It does not establish a sharp rate as s grows with k . Its constants are intentionally inefficient. In the asymptotic corollary, s, C, η are fixed first; no exchange of these quantifiers is used.

An earlier recovered draft contained a covariance–reservoir argument. The present version rederives that argument and simplifies it: successive powers use a single trajectory; pruning is implemented by direct rejection inside the original reservoir; changing one column is treated with the sharp one-Lipschitz count bound; and concentration is proved directly for sampling without replacement. The last change strengthens the former collision-error probability estimate to the exponential bound (1). The original draft is preserved separately in the bundle.

The accompanying standard-library rational tests check finite identities and small instances, including a counterexample to treating the all-hit conditional law as the ideal law. They are implementation checks, not an exhaustive verification of the universal theorem. On 12 September 2026 a separate Codex AI-agent reviewer returned PASS for the full canonical target, with no mathematical corrections. The detailed audit and the original reviewed source are preserved in the repository submission record, references/holden-tr07-2026-09-12. AI assistance was used for review and submission preparation. This is informal automated review, not external human peer review or formal verification; no Lean verification was performed. The permanent target is unchanged. The author affiliation was verified against the official Simons Foundation profile on the same date. Publication priority is not asserted.

References

- [1] H. Huang, M. Rudelson, and K. Tikhomirov, *Well-invertible column subsets of sparse matrices are rare*, arXiv:2607.05384v2 (2026), Theorem 1.3 and Conjecture 7.1. <https://arxiv.org/html/2607.05384v2>.
- [2] A. Townsend, *Open Problems in Numerical Linear Algebra*, entry TR-07, “Random column subsets of arbitrary fixed-sparsity matrices.” Canonical statement accessed 12 September 2026. <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/randomized-and-low-rank-approximation/TR-07>.